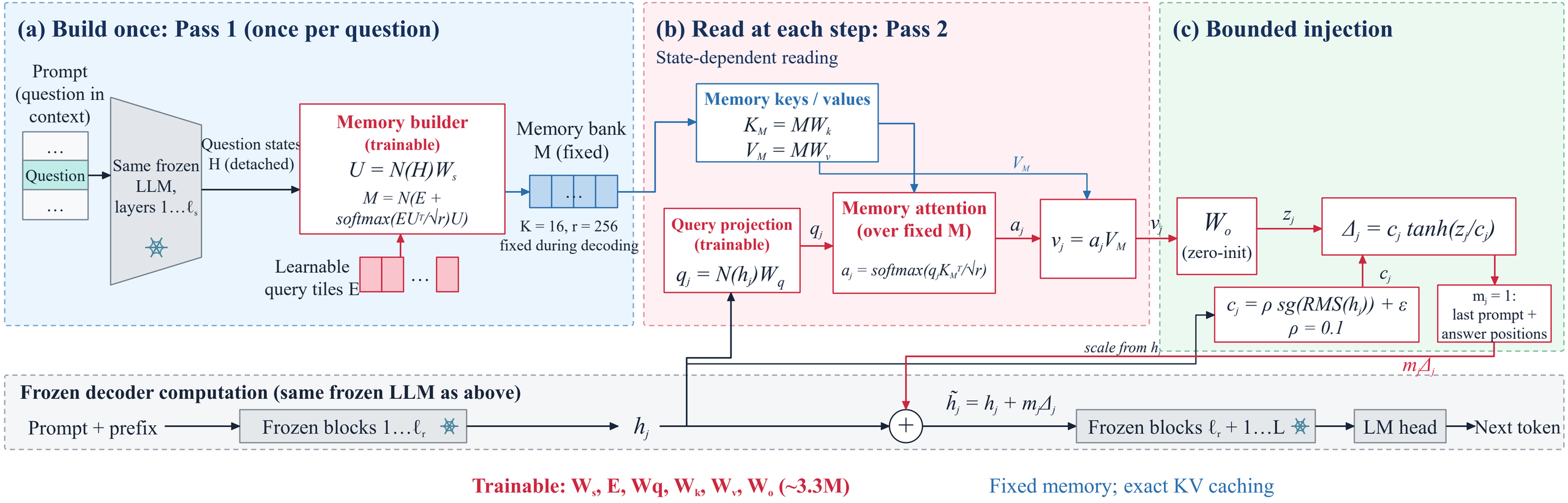

(a) Build once: Pass 1 (once per question)
Prompt (question in context)
…
Question
…
Same frozen LLM, layers 1…ℓs
Question states H (detached)
Memory builder (trainable)
U = N(H)Ws
M = N(E + softmax(EUT/√r)U)
Learnable query tiles E
Memory bank M (fixed)
K = 16, r = 256
fixed during decoding
(b) Read at each step: Pass 2
State-dependent reading
Memory keys / values
KM = MWk
VM = MWv
Query projection (trainable)
qj = N(hj)Wq
qj
Memory attention (over fixed M)
aj = softmax(qjKMT/√r)
aj
vj = ajVM
VM
vj
(c) Bounded injection
Wo
(zero-init)
zj
Δj = cj tanh(zj/cj)
cj
cj = ρ sg(RMS(hj)) + ε
ρ = 0.1
mj = 1: last prompt + answer positions
scale from hj
mjΔj
Frozen decoder computation (same frozen LLM as above)
Prompt + prefix
Frozen blocks 1…ℓr
hj
h̃j = hj + mjΔj
Frozen blocks ℓr + 1…L
LM head
Next token
Trainable: Ws, E, Wq, Wk, Wv, Wo (~3.3M)
Fixed memory; exact KV caching